

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 7_CY

Attempt : 1
Total Mark : 40
Marks Obtained : 20

Section 1 : Coding

1. Problem Statement

Maria, an online store owner, is looking to implement a pricing system that calculates the final price of products after applying discounts. She needs a program that takes the original price of a product and the discount percentage as input and computes the final discounted price. The discount is applied as a percentage of the original price. Maria wants to ensure that the final price is formatted to display exactly two decimal places.

Implement this functionality using the PriceCalculator interface and the DiscountCalculator class.

Input Format

The first line of input consists of the original price (a double value).

The second line of input consists of a discount percentage (a double value).

Output Format

The output displays the final price after the discount, adhering to the following format: "Final Price after discount: \$[final_price]".

Here, [final_price] should be replaced with the calculated final price, formatted as a currency value with two decimal places.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 100.0
10.0

Output: Final Price after discount: \$90.00

Answer

```
import java.util.Scanner;

interface PriceCalculator {
    double calculatePrice(double price, double discount);
}

class DiscountCalculator implements PriceCalculator {
    public double calculatePrice(double price, double discount) {
        return price - (price * discount / 100);
    }
}

class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        double originalPrice = scanner.nextDouble();
        double discount = scanner.nextDouble();
        PriceCalculator calculator = new DiscountCalculator();
        double finalPrice = calculator.calculatePrice(originalPrice, discount);
        System.out.printf("Final Price after discount: $%.2f\n", finalPrice); //
        Formats output to 2 decimal places
    }
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

A developer aims to create a budget management system using two interfaces, ExpenseRecorder for recording expenses and BudgetCalculator for calculating remaining budgets.

The ExpenseTracker class implements these interfaces, allowing users to input an initial budget and record expenses iteratively until entering 0.0 as a sentinel value.

The program then computes and displays the remaining budget or notifies of budget exceedance.

Example

Input

100.0

20.0 30.0 10.0 0.0

Output

Remaining budget: Rs. 40.00

Explanation

The initial budget is 100.0. Expenses of 20.0, 30.0, and 10.0 are recorded.

Remaining budget is calculated ($100.0 - 20.0 - 30.0 - 10.0 = 40.0$).

Input Format

The first line of input is the initial budget as a double-point number (double type).
The budget is a positive number.

The second line of input consists of individual expenses as double-point numbers. Each expense is separated by space.

To end the input, an expense of 0.0 is used.

Output Format

The output displays the remaining budget, formatted to two decimal places, in the following format:

If the remaining budget (double type) is non-negative, it prints "Remaining budget: Rs. [remainingBudget]".

If the remaining budget is negative, it prints "No remaining budget, You've exceeded your budget!".

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 100.0

20.0 30.0 10.0 0.0

Output: Remaining budget: Rs. 40.00

Answer

```
import java.util.Scanner;
```

```
interface ExpenseRecorder {  
    void recordExpense(double expense);  
}
```

```
interface BudgetCalculator {  
    double calculateRemainingBudget();  
}
```

```
class ExpenseTracker implements ExpenseRecorder, BudgetCalculator {  
    private double initialBudget;  
    private double totalExpenses = 0.0;
```

```
    ExpenseTracker(double budget) {  
        this.initialBudget = budget;  
    }  
}
```

```

    public void recordExpense(double expense) {
        totalExpenses += expense;
    }

    public double calculateRemainingBudget() {
        return initialBudget - totalExpenses;
    }
}

class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        double budget = scanner.nextDouble();

        ExpenseTracker tracker = new ExpenseTracker(budget);

        double expense;
        do {
            expense = scanner.nextDouble();
            tracker.recordExpense(expense);
        } while (expense != 0.0);

        double remainingBudget = tracker.calculateRemainingBudget();
        if (remainingBudget >= 0) {
            System.out.printf("Remaining budget: Rs. %.2f", remainingBudget);
        } else {
            System.out.println("No remaining budget, You've exceeded your budget!");
        }
    }
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Jeevan is developing a fitness-tracking application to monitor daily physical activity.

The application incorporates a FitnessTracker class that implements two

interfaces: StepCounter for tracking the number of steps taken and CalorieCalculator for estimating total calories burned based on total steps.

Jeevan needs your help creating a program.

Note

The calorie calculation formula is: $\text{Total caloriesBurned} = (\text{total steps} / 100.0) * 20.0$.

Input Format

The first line of input is an integer n , representing the number of days Jeevan wants to input data.

The second line consists of space-separated integers, representing the number of steps Jeevan took on each day.

Output Format

The first line of output prints: "Total Steps: <totalSteps>", where '<totalSteps>' is the sum of steps (integer) taken over ' n ' days.

The second line prints: "Calories Burned: <caloriesBurned>", where '<caloriesBurned>' is the estimated total calories (double-point number) burned based on the total steps taken rounded off to two decimal places.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3

340 234 987

Output: Total Steps: 1561

Calories Burned: 312.20

Answer

```
import java.util.Scanner;

interface StepCounter {
    int countTotalSteps(int[] steps);
}
```

```
}
```

```
interface CalorieCalculator {  
    double calculateCalories(int totalSteps);  
}
```

```
class FitnessTracker implements StepCounter, CalorieCalculator {  
    public int countTotalSteps(int[] steps) {  
        int total = 0;  
        for (int s : steps)  
            total += s;  
        return total;  
    }  
  
    public double calculateCalories(int totalSteps) {  
        return (totalSteps / 100.0) * 20.0;  
    }  
}
```

```
class Main  
{
```

```
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);
```

```
        FitnessTracker tracker = new FitnessTracker();
```

```
        int n = scanner.nextInt();
```

```
        for (int i = 0; i < n; i++) {  
            int steps = scanner.nextInt();  
            tracker.countSteps(steps);  
        }
```

```
        int totalSteps = tracker.getTotalSteps();  
        System.out.println("Total Steps: " + totalSteps);
```

```
        double caloriesBurned = tracker.calculateCaloriesBurned(totalSteps);  
        System.out.printf("Calories Burned: %.2f%n", caloriesBurned);
```

```
        scanner.close();  
    }
```

```
}
```

Status : **Wrong**

Marks : 0/10

4. Problem Statement

Alex and Bob are designing a control system for household appliances, and one of the appliances is a washing machine. You want to create a program to help them that models the washing machine as a motor and calculates its electricity consumption based on its capacity.

Define an interface named Motor with the following methods:

`void run()` `double consume(double capacity)`

Create a class called WashingMachine that implements the Motor interface.

In the WashingMachine class:

Implement the `run()` method to print "Washing machine is running." Implement a `consume()` method to print "Washing machine is consuming electricity." Implement the `consume(double capacity)` method to calculate the electricity consumption (in kWh) of the washing machine based on its capacity. The formula for electricity consumption is $(\text{capacity} * 0.05)$.

Input Format

The input consists of a double value representing the capacity of the washing machine in kW.

Output Format

The first line of output prints "Washing machine is running."

The second line prints "Washing machine is consuming electricity."

The third line prints "Electricity consumption: X kWh" where X is a double value, rounded off to two decimal places, representing the electricity consumption.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2.5

Output: Washing machine is running.

Washing machine is consuming electricity.

Electricity consumption: 0.13 kWh

Answer

-

Status : Skipped

Marks : 0/10